

Hypothesis Testing



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DATE

⇒ Confidence Intervals provide us with an Estimation of where the Parameters are located.

But, While making an Decision, a Yes or No answer is needed.

Approach to use for getting the Answer is Hypothesis Testing.

⇒ Four Steps in Data Driven Decision Making :

1) Formulate a Hypothesis

2) Find the right Test for Hypothesis.

3) Execute the Test.

4) Make a Decision based on the Result.

⇒ Hypothesis :->

A Hypothesis is an Idea that can be Tested.

It is a Supposition or Proposed explanation made on the basis of limited evidence as a Starting Point for Further Investigation.

① There are Two Hypotheses that are made:

1) Null Hypothesis :

Denoted by H_0 .

2) Alternative Hypothesis :

Denoted by H_1 or H_a .

The Null Hypothesis is the one to be Tested and The Alternative Hypothesis is everything else.



=> Null Hypothesis (H_0) Alternative Hypothesis (H_1 or H_a)

→ The Null Hypothesis is the hypothesis to be Tested. The Alternative Hypothesis is the Change or Innovation that is contesting the Status-quo.

→ It is Everything that was believed until now that we are contesting with our tests. Usually the Alternative is our own opinion.

→ The concept of Null is similar to : Innocent Until Proven Guilty. We assume Innocence until we have enough evidence to prove the guilt. If Null is what was generally believed, then the act of Performing a Test shows we have doubt about Truthfulness of Null.



E.g.

(Q1) Person A Says mean salary
of Data Scientist is 1,25,000.

But, you ~~do~~ doubt that and
want to Test ~~is~~ that.

∴ Null Hypothesis H_0 :

$$\mu(\text{salary}) = 1,25,000$$

Alternative Hypothesis H_a or H_2 :

$$\mu(\text{salary}) \neq 1,25,000$$

(Q2) Person B says that mean salary
of a Data scientist is more
than 1,25,000.

You want to Test it.

$$\therefore H_0: \mu(\text{salary}) > 1,25,000$$

$$H_a \text{ or } H_2: \mu(\text{salary}) \leq 1,25,000.$$



$\Rightarrow$ Level of Significance and Types of Tests :

① Level of Significance (α) :

The probability of Rejecting a Null Hypothesis that is True, the Probability of Making this Error.

Common Significance level (α)
Values :

0.10 , 0.05 , 0.01 .

② Types of Tests :

When Testing, there are Two Decisions that can be made :

1) To Accept the Null Hypothesis,

2) To Reject the Null Hypothesis.

1) To Accept the Null Hypothesis means that there is not Enough Data to Support the

Change or the Innovation brought by the Alternative Hypothesis.

(2) To Reject the Null means that there is Enough Statistical Evidence that The Status-quo is not Representative of the Truth.

→ Two-Sided or Two-Tailed Test:

Used when the Null Hypothesis contains an Equality ($=$) or Inequality Sign ($\neq$).

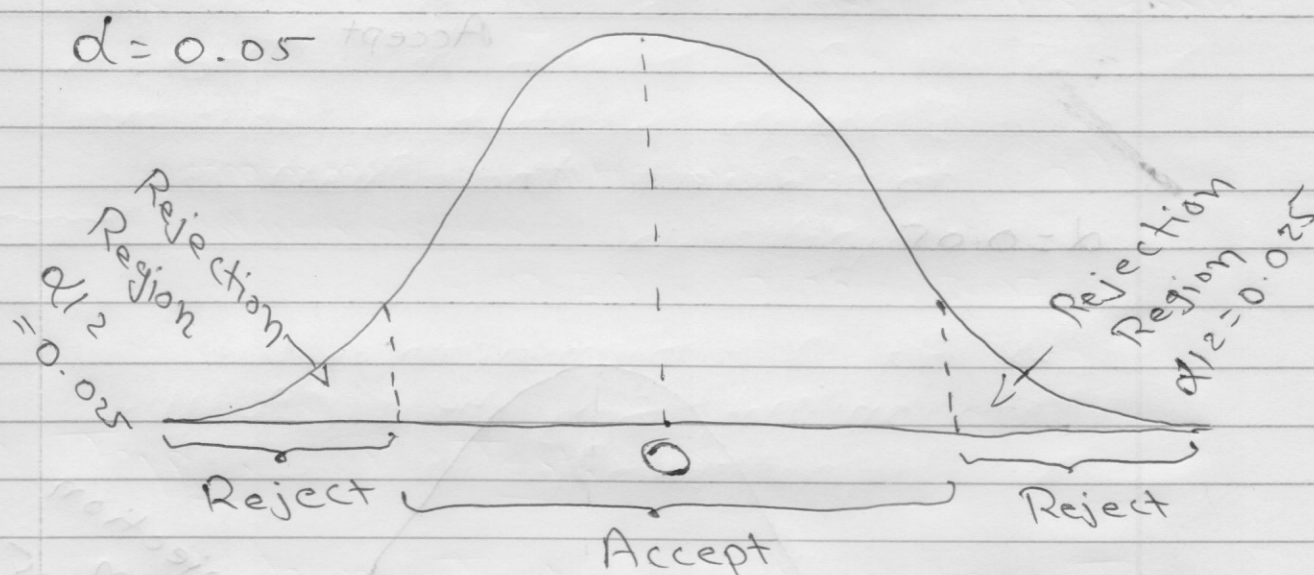
Graphically, the Tails of the Distribution show when we Reject the Null Hypothesis. ('Rejection Region').

Everything which Remains in the Middle is the 'Acceptance Region'.



The Logic is :

If the Observed Statistic is too far away From 0 (depending on the Significance level), we reject the Null Hypothesis. Otherwise, we Accept it.

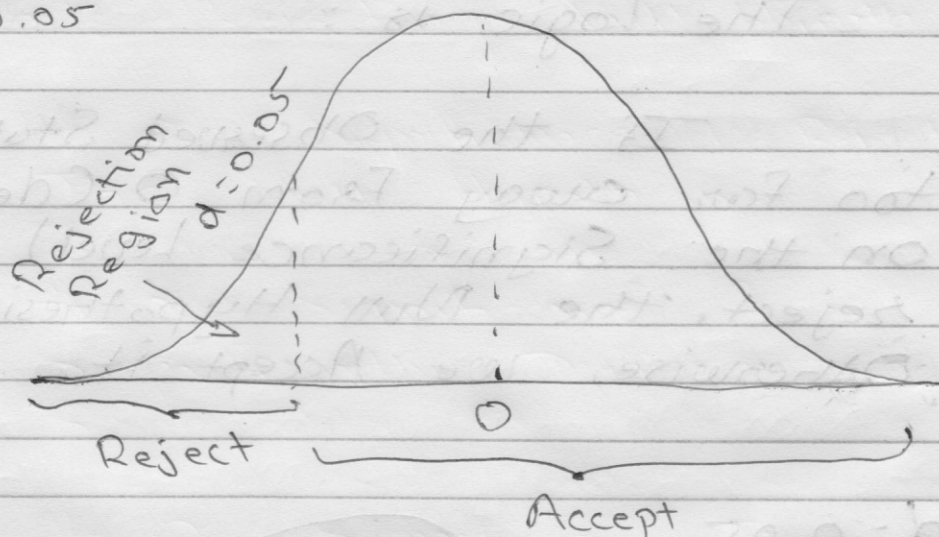


→ One-sided or One-Tailed Test :

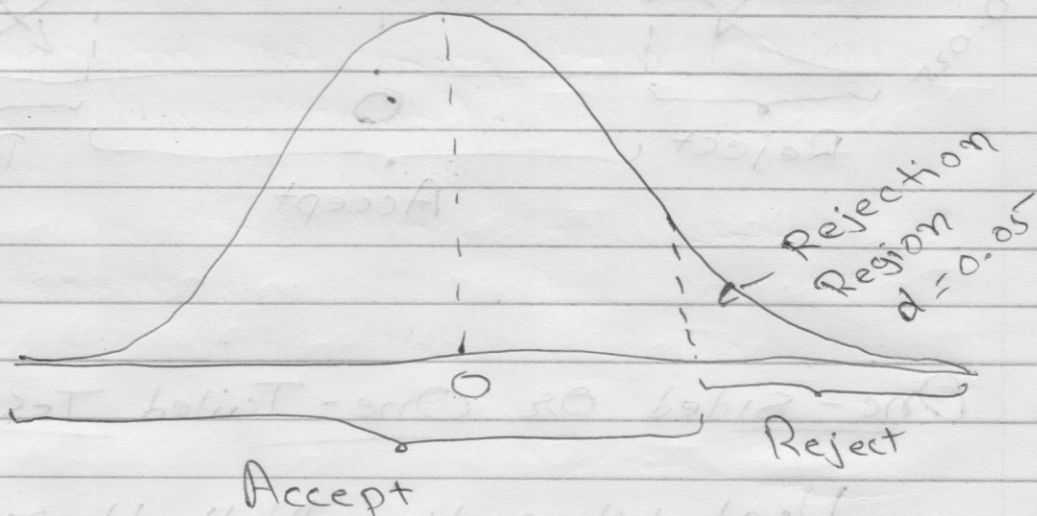
Used when the Null Hypothesis doesn't contain Equality or Inequality signs and Contains ($<$, $>$, $\leq$, $\geq$).



$$\alpha = 0.05$$



$$\alpha = 0.05$$





① Different Ways of Reporting The Result of Hypothesis Testing:

↳ While Accepting Null Hypothesis :

↳ At $\alpha\%$ significance, we Accept The Null Hypothesis.

or

↳ At $\alpha\%$ Significance, A is not Significantly Different than B.

or

↳ At $\alpha\%$ Significance, there is not Enough Statistical Evidence to Reject the Null Hypothesis.

or

↳ At $\alpha\%$ Significance, we cannot Reject The Null Hypothesis.



↳ While Rejecting Null Hypothesis :

↳ At $\alpha\%$ Significance, We Reject The Null Hypothesis.

↳ At $\alpha\%$ Significance, A is Significantly Different than B.

↳ At $\alpha\%$ Significance, There is Enough Statistical Evidence to Reject the Null Hypothesis.

↳ At $\alpha\%$ Significance, We Can Not Say that (Null Hypothesis).



$\Rightarrow$ Statistical Errors in Hypothesis Testing:

In General, there are two Types of ~~Err~~ Errors we can make while testing:

Type I Error : (False Positive)

Type II Error : (False Negative)

⊙ Type I Error :

Rejecting a True Null Hypothesis.

It is also called False Positive.

Probability of Making this Error is α , the Level of Significance.

① Type 2 Error:

Accepting a False Null Hypothesis.

It is also called False Negative.

Probability of Making this Error is Denoted by ' β '.

' β ' depends mainly on Sample Size and Magnitude of Effect.

Probability of Rejecting a False Null Hypothesis is ' $1 - \beta$ '.

As, Rejecting the False Null Hypothesis is ~~the~~ Usually the Researcher's goal, ~~$1 - \beta$~~
' $1 - \beta$ ' is called Power of the Test.



The Truth

		Ho is True	Ho is False
<u>Ho</u> (Status quo)	Accept	All good.	Type 2 Error (False Negative)
	Reject	Type 1 Error (False Positive)	All good.

$\Rightarrow$ P-Value :

Most common way to Test Hypotheses.

① The p-value is the Smallest Level of Significance at which we can still Reject the Null Hypothesis, given the Observed Samples Statistics.

② Rule : ~~You~~ S

After choosing a Significance level (α) ,

you should Reject a Null Hypothesis if $p\text{-value} < \alpha$.

e.g. If $p\text{-value} = 0.0001$

Test at 90% Significance Level :

$$\alpha = 0.1$$

$$\therefore 0.0001 < 0.1$$

$\therefore$ Reject Null Hypothesis

~~Test at 95~~

Test at 95% Significance Level:

$$\alpha = 0.05$$

$$\therefore 0.0007 < 0.05$$

$\therefore$ Reject Null Hypothesis.

Test at 99% Significance Level:

$$\alpha = 0.01$$

$$\therefore 0.0007 < 0.01$$

$\therefore$ Reject Null Hypothesis.

① p-value For one-sided Test:

~~$$p\text{-value} = 1 - \frac{\text{Sum of Row \& Column of}}{\text{Sum of Row \& Column of}}$$~~

$$p\text{-value} = 1 - \text{Value from}$$

Z-statistics or

t-statistics table

Corresponding to

Row & Column

of Statistic we

got from Test.

E.g. We got Z-Statistic
Value = 2.12 From
Test.

∴ Looking at Z-statistics table
Row : 2.1

Column : 0.02

∴ Value = 0.983

∴ p-value = 1 - 0.983
~~= 0.017~~

∴ p-value = 0.017

① p-value For Two-Sided Test :

$$p\text{-value} = 2 - \left[\begin{array}{l} \text{Value from} \\ \text{Z-Statistics} \\ \text{or t-Statistics} \\ \text{Table corresponding} \\ \text{to Row and} \\ \text{Column of} \\ \text{Statistic we} \\ \text{got from Test} \end{array} \right] \cdot 2$$



e.g. We got Z-statistic
value = 2.12 from Test.

$\therefore$ Looking at Z-statistics
table

Row : 2.1

Column : 0.02

$\therefore$ Value = 0.983

$$\begin{aligned}\therefore p\text{-value} &= (1 - 0.983) \cdot 2 \\ &= 0.034\end{aligned}$$

⊙ Notable p-values :

0.000 : This indicates that we
Reject the Null Hypothesis
at all Significance levels.

0.05 : Often called 'Cut-off' line.

If p-value is higher than
0.05, we should Accept the
Null.

⇒ Hypothesis Testing in Different Situations :->

○ On Single Population

○ On Multiple Populations :-

1.1) On Single Population :

1.1.1) With known Variance :

Test for the Mean, population Variance known

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$\bar{x}$ = mean of Sample

μ_0 = Mean from Null Hypothesis

σ = Population Variance

n = Sample Size.



Ex:

Null Hypothesis H_0 : $\mu_0 = 1,13,000$

Alternative Hypothesis H_1 :

$\mu_0 \neq 1,13,000$

$\therefore$ It's a Two-Sided Test.

Also, Sample Mean $\bar{x} = 1,00,200$

Population std. $\sigma = 15,000$

~~Standard~~ , 'std' =

Sample size = 30

$$\therefore \text{Standard Error } \frac{\sigma}{\sqrt{n}} = \frac{15,000}{\sqrt{30}}$$

$$= 2739.$$

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{1,00,200 - 1,13,000}{2739}$$

$$\therefore Z = -4.67$$

Note : The Uppercase 'Z' and lowercase 'z' are different.

Z : Standardized Variable Associated with the Test. ~~called~~

Called as Z-Score.

z : One from the table 'z-table'.

Called as Critical Value.

Now, selecting significance level of 95% $\Rightarrow$

$$\alpha = 0.05$$

As it is two sided Test, we will check for value of $z_{\alpha/2} = z_{0.025}$ from table.



$$\therefore 1 - \alpha = 1 - 0.05$$

$$= 0.95$$

$$50.0 = 10$$

$$1 - 0.025 = 0.975$$

In z-statistics table, For value

$$0.975 : \text{Row} = 1.9$$

$$\text{Column} = 0.06$$

$$\therefore \text{Critical value} = 1.96$$

Now, Comparing Absolute Value of Z-Score to Absolute Value of critical value.

$$\therefore 4.67 > 1.96$$

$$(Z)$$

$$(z)$$

Decision : Reject Null Hypothesis if
Rule $|Z\text{-Score}| > |\text{critical value}|$

$\therefore$ At 5% Significance level, We can Reject Null Hypothesis.



Selecting Significance level

of 99% $\Rightarrow$

$$\alpha = 0.01$$

Two-Sided Test $\Rightarrow \alpha/2 = 0.005$

$$\therefore 1 - 0.005 = 0.995$$

In z-statistic table, For Value

$$0.995 \Rightarrow \text{Row} = 2.5$$

$$\text{Column} = 0.08$$

$$\therefore \text{Critical Value} = 2.58$$

$$\therefore |Z\text{-Score}| = 4.67 > \text{Critical Value} = 2.58$$

$\therefore$ At 99% Significance level,
We can Reject Null Hypothesis.



1.2 With Unknown Variance : T

Test for the mean, Population Variance Unknown = T

$$T = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$\bar{x}$ = Sample Mean

μ = Population Mean

s = Sample Variance

n = Sample Size

Ex:

Null Hypothesis H_0 : $\mu \leq 40\%$

Alternative Hypothesis H_1 : $\mu > 40\%$

Sample Mean $\bar{x} = 37.70$

Sample standard deviation $s = 13.74$

Sample size = 20

$$\begin{aligned} \therefore \text{Standard Error} &= \frac{s}{\sqrt{n}} = \frac{13.74}{\sqrt{20}} \\ &= 4.34 \end{aligned}$$



$$\therefore T = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} = \frac{37.70 - 40}{4.34}$$

$$\therefore T = -0.53$$

Now, for Significance level of 95%.

$$\alpha = 0.05$$

It is One sided Test.

Degress of Freedom = $n - 1$

$$= 9$$

$$\therefore T_{9, 0.05} = 1.833 \text{ (From } t\text{-Statistic table)}$$

(Row = 9
Column = 0.05)

$$\therefore t = 1.833$$

$$\therefore |T| < |t|$$

$\therefore$ At 95% Significance level,
we accept the Null Hypothesis.



Q2) On Multiple Populations:

Q2.1) Test For Mean, Dependent Samples:

$$T = \frac{\bar{d} - \mu_0}{S_d / \sqrt{n}}$$

$\bar{d}$ = Difference of Mean

μ_0 = Hypothesized mean

S_d = Std. deviation of difference

n = Sample Size.

Ex: Magnesium Pills:

Before	After	Difference (B-A)
2	1.7	0.3
1.4	1.7	-0.3
1.3	1.8	-0.5
1.1	1.3	-0.2
1.8	1.7	0.1
1.6	1.5	0.1
1.5	1.6	-0.1
0.7	1.7	-1
0.9	1.7	-0.8
2.5	2.4	-0.9



$$\text{Sample mean} = -0.33$$

$$\text{Sample Std} = 0.45$$

$$\text{Sample size} = 20$$

$$\therefore \text{Standard Error} = \frac{S_d}{\sqrt{n}}$$

$$= \frac{0.45}{\sqrt{20}} \\ = 0.1$$

Null Hypothesis $H_0: \mu_B \geq \mu_A$
(Before) (After)

$$\therefore \mu_B - \mu_A \geq 0$$

$$H_0: D_0 \geq 0$$

Alternative Hypothesis $H_1: \mu_B < \mu_A$

$$\therefore \mu_B - \mu_A < 0$$

$$H_A: D_0 < 0$$



$$T = \frac{\bar{x} - \mu_0}{\frac{S_d}{\sqrt{n}}} \quad (5.5)$$

$$= \frac{-0.33 - 0}{0.74} = -0.44$$

$$\therefore T = -2.29$$

p-value = 0.024 ; Looking For
value 2.29
in t-statistic

$\therefore$ At significance level 95% :

$$\alpha = 0.05$$

$$p\text{-value} < \alpha$$

$\therefore$ At 95% Significance Level,
we can reject Null Hypothesis.

At Significance level 99%:

$$\alpha = 0.01$$

$$\therefore p\text{-value} > \alpha$$

$\therefore$ At 99% Significance Level, we can
not reject Null Hypothesis.

2.2 Test for Two means, Independent Samples, known Population Variance:

$$Z = \frac{\bar{x} - \mu_0}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}}$$

$\bar{x}$ = Difference of Sample Means

μ_0 = Hypothesized Difference of means

σ_x, σ_y = ~~Std~~ Population Standard Deviations

n_x, n_y = Sample Sizes

Ex: Difference between Average Engineering Grades and Average Management Grades is 4% ($\mu_E - \mu_M = 4\%$).

Null Hypothesis H_0 : $\mu_E - \mu_M = -4\%$.

Alternative Hypothesis H_1 : $\mu_E - \mu_M \neq -4\%$.



	Engineering	Management	Diff
Size	200	70	-
Mean	58.1	65.1	-7.1
Population Std	10.1	6.1	1.23

$$\sqrt{\frac{6e^2}{n_e} + \frac{6m^2}{n_m}} = \sqrt{\frac{100}{100} + \frac{36}{70}}$$
$$= 1.23$$

$$Z = \frac{\bar{x} - \mu_0}{\sqrt{\frac{6m^2}{n_m} + \frac{6e^2}{n_e}}}$$

$$= \frac{-7.1}{1.23} = -5.77$$

$$\therefore Z = -2.44$$

$\therefore p\text{-Value} = 0.015$ (Look for value in z-table corresponding to Row: 2.4, Column: 0.04)

$$\therefore \text{Value} = 0.9925$$

(Two-Sided test)

$$\therefore p\text{-value} = (1 - 0.9925) \cdot 2 = 0.015$$



At 95% Significance level:

$$\alpha = 0.05$$

$$\therefore p\text{-value} < \alpha$$

$\therefore$ At 95%, we can Reject Null Hypothesis.

(2.3) Test for Two means, Independent Samples, Unknown Population Variances (Assumed to be equal):

$$T = \frac{(\bar{x} - \bar{y}) - \mu_0}{\sqrt{\frac{S_p^2}{n_x} + \frac{S_p^2}{n_y}}}$$

$$S_p^2 = \frac{(n_x - 1) \cdot S_x^2 + (n_y - 1) \cdot S_y^2}{n_x + n_y - 2}$$

$$n_x + n_y - 2$$



$\bar{X}, \bar{Y}$ = Sample Means

μ_0 = Difference between

hypothesized True means

S_p^2 = Pooled Variance

n_x, n_y = Sample Sizes

Ex:

	A	B	(Hypotheses in back)
Mean	3.94	3.25	
Std. deviation	0.18	0.27	
Sample Size	10	8	

Pooled Variance $S_p^2 =$

$$\frac{(10-1) \cdot (0.18)^2 + (8-1) \cdot (0.27)^2}{(10+8-2)}$$

$$\therefore S_p^2 = 0.05$$

$$\therefore \text{Standard Error} = \sqrt{\frac{S_p^2}{n_a} + \frac{S_p^2}{n_b}}$$

$$= \sqrt{\frac{0.025}{10} + \frac{0.025}{8}}$$

$$= \sqrt{\frac{0.05}{10} + \frac{0.05}{8}} = 0.17$$



$$\therefore T = \frac{(\bar{x} - \bar{y}) - \mu_0}{\text{Standard Error}}$$

$$= \frac{(3.94 - 3.24) - 0}{0.77}$$

$$\therefore T = 6.53$$

~~$\mu_0 = 0$ because~~

Null Hypothesis : $H_0 : \mu_A - \mu_B = 0$

Alternative Hypothesis $H_1 : \mu_A - \mu_B \neq 0$

Two tailed Test.

① Degrees of Freedom = $n_A + n_B - 2$

$$= 32 + 32 = 64$$

For significance level of 95 %

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$F_{0.025, 64, 64} = 70.01$$



p-value can't be found in t-statistic table as $T = 6.53$ is very High.

Generally, For z & T , a value higher than t is extremely Significant and Null Hypothesis should be Rejected.

From online calculators, we can find p-value for this Case =

0.000002

$$\therefore \cancel{P < \alpha} \quad P\text{-value} < \alpha$$

$\therefore$ We can Reject Null Hypothesis at all levels of Significance.